

Pretraining Domain and Label Efficiency for Dark-Vessel Detection in SAR: A Controlled ViT–CNN Study

John Roth Kyle Wagner

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Abstract

Dark vessels lack a matched Automatic Identification System report, which makes them harder to identify and monitor. Spaceborne synthetic aperture radar (SAR) observes through cloud and darkness, but human-verified examples remain scarce. We measure how pretraining source changes label efficiency on xView3-SAR and whether the ranking persists across backbone families. Two approximately size-matched tracks use ViT-B/16 and ConvNeXt-V2-Base. Within each track, random, ImageNet, optical remote-sensing, and SAR initializations share one detector, training recipe, scene split, and seed. We fine-tuned four nested budgets from 12 to 111 training scenes (10%, 25%, 50%, and 100%), for 32 controlled cells. A seeded three-channel stem turned the ten-band optical CNN checkpoint’s apparent negative transfer into a gain. At 12 scenes, every pretrained arm beat its random floor. On the once-scored 16-scene test split, SAR led optical in the CNN track at 12 to 56 scenes but trailed at 111, the ViT track showed no stable order, and two SAR curves failed the predeclared monotonicity gate. On the once-only 50-scene human-verified set, F1 fell to 0.409–0.574 and no cell exceeded 0.228 dark-vessel recall, yet pretraining still beat random initialization at 12 scenes. All values are seed-0 point estimates.

1 Introduction

Maritime monitoring systems often identify vessels by matching detections with Automatic Identification System reports. Vessels without a contemporaneous match are commonly called *dark vessels*. Sentinel-1 synthetic aperture radar (SAR) can observe them at night and through most weather conditions [7]. These detections support maritime safety, fisheries monitoring, and investigations of activity that may involve sanctions evasion. Detection remains difficult. Vessels occupy few pixels in a large scene, while sea clutter and coastal structures also produce strong responses. Human verification adds further cost. The xView3-SAR benchmark combines ground-range-detected imagery, georeferenced point annotations, and geographic matching to make this problem measurable [3]. Pretraining can reduce the demand for verified target labels, but several source domains are plausible. ImageNet supplies generic visual features. Optical remote-sensing imagery adds overhead-scene structure. SAR pretraining also shares the target sensor modality. A domain match alone may not determine transfer quality because architectures, objectives, and source corpora also differ. Released checkpoints do not support a fully crossed domain-by-objective experiment. A comparison that also changes the detector or optimization recipe would make the source effect harder to interpret.

We compare two architecture-matched tracks (Fig. 3, Appendix B). One uses ViT-B/16 and the other ConvNeXt-V2-Base. Each track contains random, ImageNet, optical, and SAR initializations. Within a track, every arm uses the same point detector, three-channel input, optimizer, schedule, nested scene subsets, and seed. This design yields 32 cells: eight arms at four label budgets. We ask whether each pretrained arm beats its track-matched random floor as labels increase. We then compare SAR with optical pretraining inside each architecture. Once-scored held-out results test whether those rankings survive model selection on eight development scenes. The results do not support one backbone-independent ordering of pretraining sources. Every pretrained arm beats random initialization with 12 training scenes. SAR leads optical in the CNN track only while labels are scarce, and the ViT track shows no stable order. Two SAR test curves fail the predeclared monotonicity gate. A once-only human-verified evaluation shows that every detector misses most dark and near-shore vessels. The detector is intentionally conventional. The contribution is a controlled comparison of released checkpoints, supported by one label contract and checkpoint-bound operating thresholds.

Table 1: Frozen scene-level data protocol and annotation support under the corrected ground-truth contract. The 50 verified scenes served one once-only final evaluation; their row gives the vessel support scored there.

Partition	Scenes	Vessel H/M	Non-vessel H/M	LOW ignore
Training-time dev subset	8	517	107	118
Development (full)	23	1,479	804	441
Test	16	1,165	420	325
Verified final	50	8,642	3,644 dark, 2,686 near shore	

2 Dataset and Evaluation Protocol

Study cohort. We use xView3-SAR Sentinel-1 ground-range-detected scenes with VH and VV polarizations at 10m pixel spacing [3]. Scene identifiers define every split. A frozen seed-0 cohort contains 150 study scenes: 111 train, 23 development, and 16 test. The split is stratified by coarse geographic region and shoreline presence. A separate, disjoint set of 50 near-shore, human-verified scenes stayed sealed until the training cohort and the test results were frozen, and was then scored once. The four training budgets are exact nested sets of 12, 28, 56, and 111 scenes:

$$S_{10} \subset S_{25} \subset S_{50} \subset S_{100}. \quad (1)$$

Every arm uses the same scenes at a given budget. Preprocessing converts each scene into a fixed three-channel decibel-domain tensor [VH, VV, VH − VV]. Training scenes alone supply the normalization statistics. The pipeline cuts overlapping 800-pixel chips, then samples shared 512-pixel crops during optimization.

Ground-truth contract. Training and evaluation use the same label rules (Fig. 5, Appendix B). A row defines a positive vessel point only when `is_vessel` is true at HIGH or MEDIUM confidence. HIGH/MEDIUM non-vessels are background, and LOW-confidence rows define ignore regions. Table 1 lists the annotation support under this contract. An audit receipt binds these counts to the frozen split file by SHA-256. Evaluation stops if regenerated counts differ. Training never stages test or human-verified labels. Figure 2 compares one full scene with a single chip and maps the scene centroids of the frozen splits.

Selection and scoring. The frozen scorer greedily matches predictions to ground truth within 200m, then aggregates precision, recall, and F1 across scenes. Every five epochs, each run evaluates the same eight development scenes. A threshold sweep selects the checkpoint-bound operating point with the highest development F1. The run record binds its epoch, threshold, and counts to the checkpoint SHA-256. Held-out stages apply that stored threshold without retuning, first to the 16 test scenes and then once to the human-verified set. No scene crosses split boundaries. Sentinel-1 revisits still place some held-out scenes near training scenes, so absolute performance measures in-region rather than geographically independent generalization.

3 Related Work

xView3 evaluates georeferenced vessel points in Sentinel-1 SAR rather than classifying bounding boxes [3]. A CenterNet-style heatmap matches both the point annotations and the distance-based metric without requiring synthetic box extents [9]. We use this established detector form as a common task head.

ImageNet pretraining provides the generic transfer baseline [4]. The ViT control uses a supervised AugReg ViT-B/16 checkpoint. The CNN control uses ConvNeXt-V2-Base after fully convolutional masked autoencoding and supervised ImageNet-1K fine-tuning [8]. The remote-sensing checkpoints add two source domains. SatDINO applies DINO self-distillation to fMoW-RGB optical imagery [5]. SARMAE applies masked autoencoding to a million-scale SAR corpus [2]. Their objectives and source corpora differ, so the ViT comparison measures the released checkpoints as a whole. It cannot isolate sensing modality.

The CNN pair offers a cleaner modality contrast. Its two BigEarthNet-v2 models share the architecture, dataset family, and supervised land-cover task. One model uses Sentinel-2 optical input and the other uses Sentinel-1 SAR input [6, 1]. Comparisons across matched roles then test whether an observed within-track ranking persists after the backbone family changes. Different pretraining histories still limit that cross-track interpretation.

Table 2: Controlled initialization roles. All non-random entries are released, revision-pinned checkpoints. We discard their source task heads.

Role	ViT-B/16 track	ConvNeXt-V2-Base track
Random	fresh initialization	fresh initialization
Optical	SatDINO, DINO on fMoW-RGB	BigEarthNet-v2 Sentinel-2
SAR	SARMAE, MAE on SAR-1M	BigEarthNet-v2 Sentinel-1
ImageNet	supervised AugReg	FCMAE + supervised

4 Methods

Two matched tracks, eight arms. Table 2 lists the eight arms. Architecture and parameter count stay fixed within each track, so only the initialization changes. We transfer encoder weights for pretrained arms and discard the source task heads. Random floors use the same target architectures. The complete detectors contain 89,996,801 trainable parameters for ViT and 93,988,865 for CNN.

Input adaptation for the BigEarthNet-S2 checkpoint. The Sentinel-2 checkpoint expects ten optical bands. An earlier cohort kept three band kernels of its first convolution and scaled them by 10/3, which raised the first-layer weight norm 5.85-fold and left the arm at or below random. Here the target model keeps its seeded three-channel first convolution and loads every other pretrained tensor. All 32 cells were then retrained from initialization under one code version.

Shared point detector. Every arm predicts a stride-4 vessel-presence heatmap from the same three-channel input (Fig. 4, Appendix B). Gaussian targets with a standard deviation of two output pixels mark positives. LOW-confidence annotations define ignore neighborhoods. Training uses the penalty-reduced focal objective from CenterNet-style point detection [9]. The ViT adapter receives stride-16 features. The CNN adapter receives stride-32 features and adds one learned upsampling stage. Both adapters terminate in the same 256-channel head. At inference, local peaks above a 0.05 candidate floor map to scene coordinates. Distance-based non-maximum suppression at 120 m removes duplicates. The tiler processes 512-pixel windows at stride 384.

Uniform fine-tuning recipe. One PyTorch Lightning module, data module, and trainer configuration serve all cells. AdamW uses a learning rate of 10^{-4} , weight decay 0.05, layer-wise decay 0.65, five warm-up epochs, and cosine decay for at most 50 epochs. Early stopping has a patience of four development evaluations. Every cell uses micro-batch 16, accumulation 1, seed 0, and full-encoder fine-tuning. The campaign ran on one eight-GPU NVIDIA H100 node. Each cell executed as a single-GPU process with Lightning 32-true. CUDA matmul and cuDNN TF32 remained disabled. Every arm and label budget uses the same hyperparameters.

Registered contrasts. For track t , role r , and fraction f , let $F_{r,f}^{(t)}$ denote development F1 at that cell’s checkpoint-bound operating threshold. The analysis reports transfer over the track-matched random floor,

$$\Delta_{r,f}^{(t)} = F_{r,f}^{(t)} - F_{\text{random},f}^{(t)}, \quad (2)$$

and the within-track SAR-optical contrast,

$$\Delta_{\text{SAR-opt},f}^{(t)} = F_{\text{SAR},f}^{(t)} - F_{\text{optical},f}^{(t)}. \quad (3)$$

We assess architecture generality from agreement in the sign and pattern of Eq. (3) across tracks. We do not pool the tracks into one effect estimate because their architectures and checkpoint histories differ.

5 Results and Analysis

Execution summary and reporting contract. All 32 cells trained from initialization on eight NVIDIA H100 80GB HBM3 GPUs in FP32 (code 06b2e61d) and used 567.6 GPU-hours. Table 3 reports corrected development-selection F1. Each cell selects its checkpoint and operating threshold on the same eight-scene subset. These scores therefore include model-selection optimism and are not held-out estimates. Figure 1 shows the four-point curves, and Fig. 6 shows their training histories in the appendix. The evidence generator

Table 3: Development-selection F1 for all 32 cells at each cell’s checkpoint-bound operating threshold. Budgets are fractions of the 111 training scenes (scene counts in parentheses). Per-cell thresholds, best epochs, and held-out slots appear in Appendix A.

Track	Initialization	10% (12)	25% (28)	50% (56)	100% (111)
ViT-B/16	Random	0.7917	0.8569	0.8935	0.8996
	Optical	0.8933	0.8854	0.9201	0.9222
	SAR	0.8857	0.8624	0.9184	0.9293
	ImageNet	0.8980	0.8765	0.9234	0.9367
ConvNeXt-V2-B	Random	0.7992	0.8167	0.8716	0.9002
	Optical	0.8371	0.8619	0.8826	0.9045
	SAR	0.8762	0.8868	0.8998	0.9068
	ImageNet	0.8889	0.8962	0.9268	0.9347

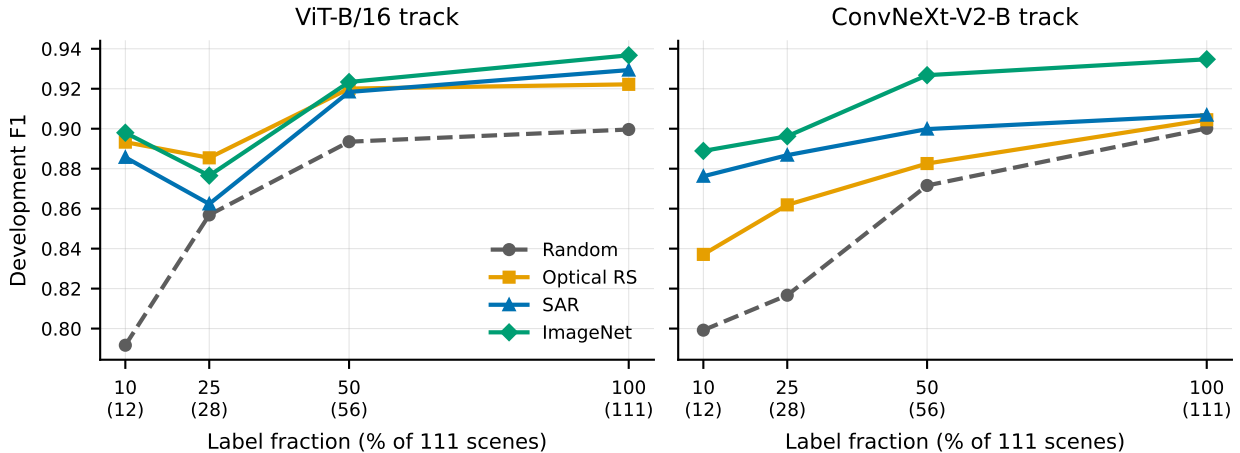


Figure 1: Development-F1 label-efficiency curves, one panel per track. Solid lines are pretrained arms. The dashed gray line is the track’s random floor. Values are seed-0 point estimates. We draw no uncertainty ribbon.

reconstructs every value from the frozen cohort, checkpoint-bound completion records, and full training curves. Two cells ran their last development evaluation in a zero-step resume that did not save the curve; their markers bind the recovery record instead (Appendix D). The generator rejects an incomplete cohort or a value that disagrees with its source record.

Label-efficiency curves and transfer gains. At 12 scenes, every pretrained arm beats its random floor in both tracks. The ViT gains are +0.102 for optical, +0.094 for SAR, and +0.106 for ImageNet (Eq. (2) and Fig. 7). The CNN gains are +0.038, +0.077, and +0.090. CNN-optical now beats random at every budget because of the input adapter (Sect. 4).

At 12 scenes, ViT-ImageNet comes within 0.002 of the full-data ViT random floor (0.8980 versus 0.8996).

Source-domain contrasts and architecture generality. The SAR–optical ranking depends on the backbone (Eq. (3) and Fig. 7). On development F1, CNN-SAR leads CNN-optical at every budget, but the gap shrinks from 0.8762 versus 0.8371 at 12 scenes to 0.9068 versus 0.9045 at 111. The BigEarthNet Sentinel-1 and Sentinel-2 checkpoints share their architecture, dataset family, and supervised task, which makes this the cleanest modality comparison in the study. In the ViT track, optical leads SAR at 10%, 25%, and 50%, and SAR leads only at 100%. The scarce-label SAR–optical order therefore does not replicate across architectures. The ViT checkpoints also differ in objective and source corpus, so their ranking cannot identify a modality-only effect.

Held-out test results. All 32 cells were scored once on the frozen 16-scene test split with their development-bound thresholds and 32-true inference. The test column appears only after every result validates against the frozen cohort. Test F1 is lower than development F1 for every cell, by 0.029 to 0.142 and 0.078 on

Table 4: F1 on the 50 human-verified scenes, scored once with each cell’s development-bound threshold. Appendix A adds recall, dark-vessel recall, and near-shore F1.

Track	Initialization	10% (12)	25% (28)	50% (56)	100% (111)
ViT-B/16	Random	0.412	0.456	0.501	0.522
	Optical	0.486	0.506	0.515	0.562
	SAR	0.493	0.425	0.568	0.559
	ImageNet	0.523	0.500	0.551	0.558
ConvNeXt-V2-B	Random	0.409	0.441	0.492	0.511
	Optical	0.457	0.479	0.534	0.552
	SAR	0.501	0.493	0.528	0.478
	ImageNet	0.464	0.500	0.562	0.574

average, as selection on eight scenes leads one to expect.

Every pretrained arm still beats its random floor at 12 scenes. In the CNN track, SAR leads optical at 12, 28, and 56 scenes (0.8000 versus 0.7197 at 12) and trails it at 111 (0.8221 versus 0.8405). In the ViT track, the two arms tie at 12 scenes (0.8041 versus 0.8064), optical leads at 28, and SAR leads at 56 and 111. The full-data winner changes in one track: ViT-SAR reaches 0.8861, above ImageNet’s 0.8509, while CNN-ImageNet keeps its lead at 0.8876. On test, 28 pretrained scenes already match the full-data random floor in both tracks. The 28 cells outside the CNN-optical arm repeat the recipe and seed of an earlier cohort (code 1a82d508), yet their test F1 differs from it by up to 0.043, so we treat contrasts below about 0.04 as run-to-run variation.

The predeclared monotonicity gate failed on two SAR test curves. ViT-SAR falls from 0.8041 at 10% to 0.7715 at 25%, and CNN-SAR falls from 0.8478 at 50% to 0.8221 at 100% although its development F1 rises. Both declines exceed the allowed 0.02, so the experiment does not establish a monotone label-efficiency curve. The final evaluation proceeded with this failure disclosed, with no retraining and no threshold tuning.

Once-only human-verified evaluation. All 32 cells were scored once on the 50 human-verified scenes (Table 4). They hold 8,642 vessels, of which 3,644 are dark and 2,686 lie within 2km of shore. F1 falls to 0.409–0.574, on average 0.306 below test F1. Precision stays between 0.639 and 0.909, but recall drops to 0.286–0.434. The detectors therefore miss most verified vessels rather than over-detect. No cell exceeds 0.041 near-shore F1 or 0.228 dark-vessel recall.

The 12-scene pretraining advantage survives this shift: the six gains over random range from +0.048 to +0.111, and ViT-ImageNet at 12 scenes matches the ViT random floor at 111 (0.523 versus 0.522). CNN-SAR again leads optical only with few labels (0.501 versus 0.457 at 12 scenes, 0.478 versus 0.552 at 111). Both SAR declines from the test split recur on this set and grow larger.

6 Conclusion and Future Work

Pretraining improved label efficiency where labels are scarcest. At 12 scenes, all six pretrained checkpoints beat their random floors on development selection, on the held-out test split, and on the once-only human-verified set. No single source won across both backbones and all evaluation stages. In the CNN track, SAR pretraining leads optical pretraining with few labels, and the lead shrinks as labels grow. On test it reverses at 111 scenes. The ViT track shows no stable SAR–optical order. The input adapter mattered as much as the source domain: a seeded three-channel stem turned the ten-band BigEarthNet-S2 checkpoint from no gain into a gain at every budget, so cross-sensor studies should report the adapter rule.

The failed monotonicity gate and rerun variation of up to 0.043 test F1 limit these conclusions. The grid is a set of descriptive seed-0 point estimates, not evidence that performance improves monotonically with more labels.

The human-verified set exposes the main open problem. Every detector loses about 0.306 F1 against test, and no cell exceeds 0.228 dark-vessel recall or 0.041 near-shore F1. The training subsets are not restricted to near-shore scenes, so this result measures coastal generalization under different clutter.

Each cell uses one seed, selection uses eight scenes, and the ViT optical and SAR checkpoints differ in objective and corpus. The results compare released checkpoints under one recipe; they do not identify a universal causal effect of pretraining domain. Future work should add seeds, near-shore training scenes, a geographically independent test cohort, and checkpoints with matched pretraining objectives.

References

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A Full Evidence Inventory

Table 5 lists every cell: development-selection F1, the checkpoint-bound operating threshold, the epoch of the selected checkpoint, the epochs run, measured GPU-hours, and held-out test F1 where admitted. Table 6 lists the once-only human-verified results. The build re-derives all values from the committed cohort, markers, training curves, and runtime receipts. Dashes mean the fail-closed generator admitted no value, never zero performance.

Table 5: Per-cell evidence inventory for development selection. The test column is admitted only when all 32 immutable held-out results validate.

Cell	Budget	Dev F1	Threshold	Best ep.	Epochs	GPU-h	Test F1
ViT random	10%	0.7917	0.808	14	35	3.1	0.6926
ViT random	25%	0.8569	0.858	24	45	7.5	0.7729
ViT random	50%	0.8935	0.852	34	50	14.9	0.8091
ViT random	100%	0.8996	0.818	24	45	25.2	0.8306
ViT optical	10%	0.8933	0.901	49	50	4.2	0.8064
ViT optical	25%	0.8854	0.921	29	50	8.2	0.8371
ViT optical	50%	0.9201	0.938	34	50	14.8	0.8369
ViT optical	100%	0.9222	0.897	44	50	28.0	0.8567
ViT SAR	10%	0.8857	0.587	4	25	2.2	0.8041
ViT SAR	25%	0.8624	0.768	4	25	4.1	0.7715
ViT SAR	50%	0.9184	0.878	39	50	14.9	0.8652
ViT SAR	100%	0.9293	0.918	39	50	28.2	0.8861
ViT ImageNet	10%	0.8980	0.849	39	50	4.2	0.8129
ViT ImageNet	25%	0.8765	0.892	29	50	8.1	0.8476
ViT ImageNet	50%	0.9234	0.859	39	50	14.8	0.8589
ViT ImageNet	100%	0.9367	0.861	44	50	28.2	0.8509
CNN random	10%	0.7992	0.716	29	50	7.1	0.6571
CNN random	25%	0.8167	0.819	44	50	14.2	0.7065
CNN random	50%	0.8716	0.846	49	50	26.6	0.7803
CNN random	100%	0.9002	0.858	39	50	51.5	0.8155
CNN optical	10%	0.8371	0.825	49	50	6.8	0.7197
CNN optical	25%	0.8619	0.886	34	50	14.0	0.7505
CNN optical	50%	0.8826	0.864	39	50	26.8	0.8214
CNN optical	100%	0.9045	0.847	49	50	51.4	0.8405
CNN SAR	10%	0.8762	0.776	34	50	6.7	0.8000
CNN SAR	25%	0.8868	0.876	14	35	9.8	0.8030
CNN SAR	50%	0.8998	0.900	29	50	26.5	0.8478
CNN SAR	100%	0.9068	0.832	4	25	25.8	0.8221
CNN ImageNet	10%	0.8889	0.861	4	25	3.4	0.7925
CNN ImageNet	25%	0.8962	0.937	9	30	8.5	0.8458
CNN ImageNet	50%	0.9268	0.892	34	50	26.5	0.8734
CNN ImageNet	100%	0.9347	0.896	49	50	51.4	0.8876

The sanitized operator status snapshot from the earlier cohort’s deadline remains committed beside the evidence tree as a historical record of execution state. It does not describe the cohort reported here.

Table 6: Per-cell results on the 50 human-verified scenes, scored once with each cell’s development-bound threshold. Dark-vessel recall covers the 3,644 vessels without an AIS match; near-shore F1 covers the 2,686 vessels within 2 km of shore.

Cell	Budget	Final F1	Recall	Dark recall	Near-shore F1
ViT random	10%	0.412	0.291	0.150	0.018
ViT random	25%	0.456	0.325	0.155	0.019
ViT random	50%	0.501	0.363	0.174	0.023
ViT random	100%	0.522	0.382	0.182	0.038
ViT optical	10%	0.486	0.347	0.149	0.022
ViT optical	25%	0.506	0.377	0.166	0.023
ViT optical	50%	0.515	0.370	0.159	0.020
ViT optical	100%	0.562	0.417	0.184	0.028
ViT SAR	10%	0.493	0.368	0.189	0.031
ViT SAR	25%	0.425	0.286	0.136	0.015
ViT SAR	50%	0.568	0.425	0.197	0.027
ViT SAR	100%	0.559	0.403	0.168	0.021
ViT ImageNet	10%	0.523	0.390	0.178	0.031
ViT ImageNet	25%	0.500	0.363	0.150	0.024
ViT ImageNet	50%	0.551	0.409	0.178	0.021
ViT ImageNet	100%	0.558	0.411	0.175	0.028
CNN random	10%	0.409	0.300	0.168	0.016
CNN random	25%	0.441	0.329	0.173	0.019
CNN random	50%	0.492	0.361	0.183	0.025
CNN random	100%	0.511	0.373	0.178	0.020
CNN optical	10%	0.457	0.354	0.193	0.027
CNN optical	25%	0.479	0.363	0.199	0.020
CNN optical	50%	0.534	0.419	0.227	0.023
CNN optical	100%	0.552	0.434	0.228	0.026
CNN SAR	10%	0.501	0.398	0.194	0.041
CNN SAR	25%	0.493	0.369	0.187	0.027
CNN SAR	50%	0.528	0.392	0.179	0.020
CNN SAR	100%	0.478	0.331	0.147	0.015
CNN ImageNet	10%	0.464	0.321	0.142	0.040
CNN ImageNet	25%	0.500	0.362	0.169	0.026
CNN ImageNet	50%	0.562	0.418	0.180	0.027
CNN ImageNet	100%	0.574	0.421	0.178	0.026

B Study Data and Figures

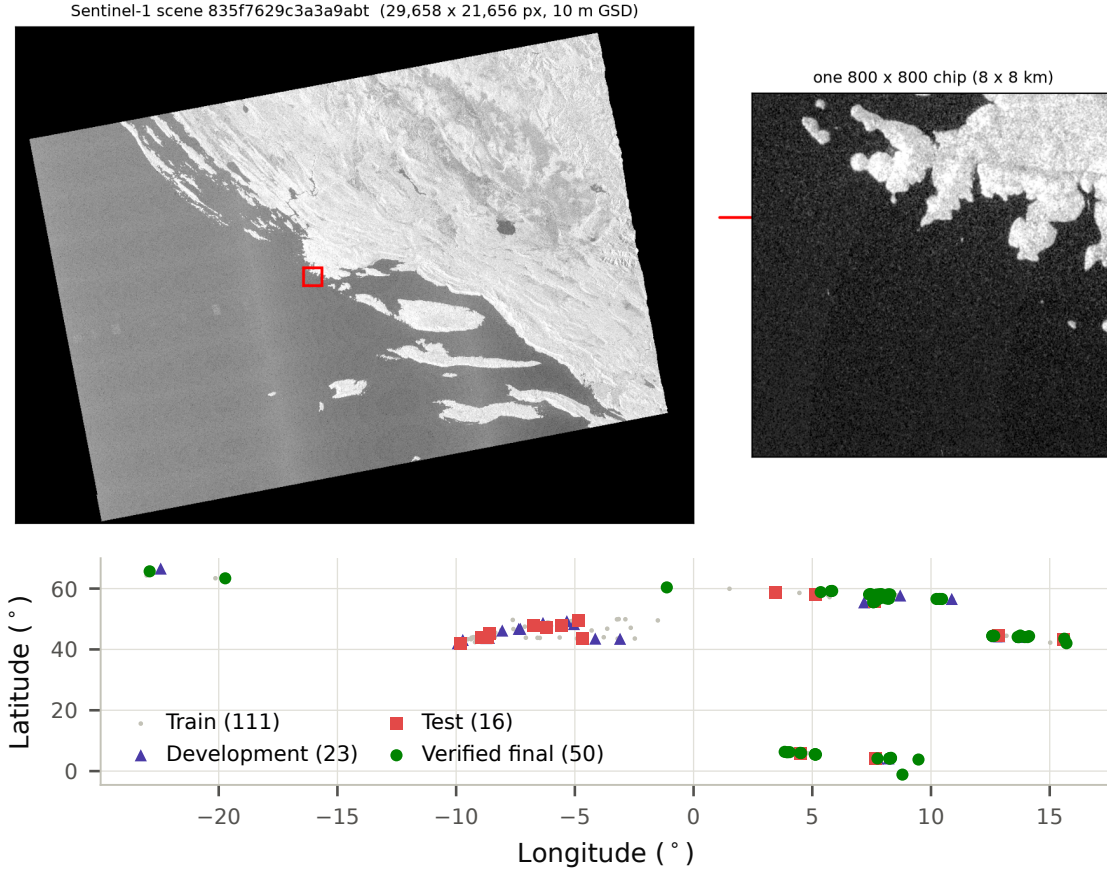


Figure 2: Study data. Top: scale context for one study scene, a full Sentinel-1 ground-range-detected scene and the single 800×800 chip (8×8 km) marked by the red box. Vessels occupy only a few pixels at this scale. Bottom: scene centroids of the frozen splits, computed from the label records. Sentinel-1 revisits place held-out scenes near training scenes, which is why Sect. 2 reports in-region generalization.

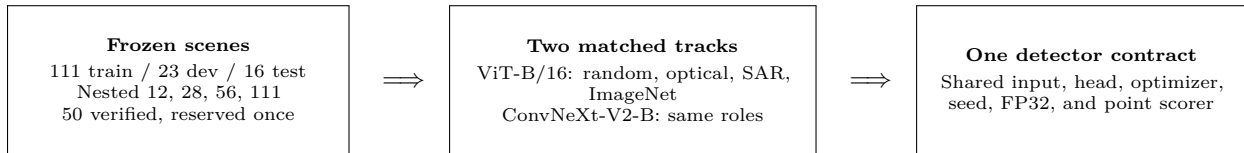


Figure 3: Controlled study design. Initialization is the only variable within each architecture track. Matched roles are compared descriptively across tracks; the human-verified set stayed sealed until the cohort and the test results were frozen, and was then scored once.

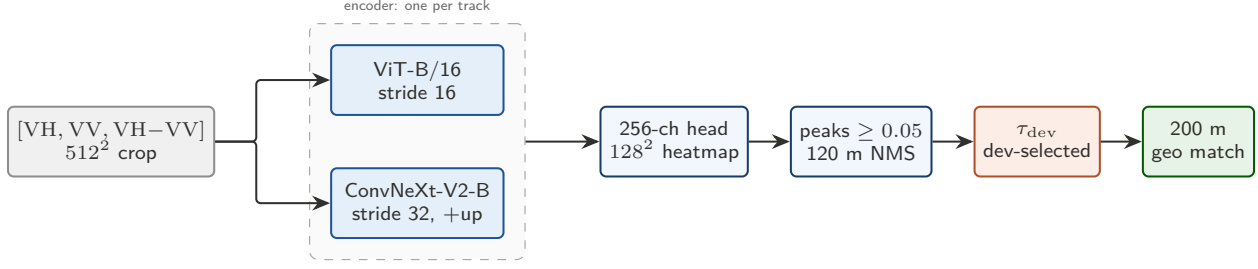


Figure 4: The shared detection path: only the encoder differs by track; one head, decode rule, dev-selected threshold, and frozen scorer serve every arm.

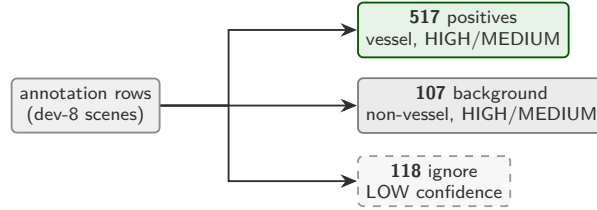


Figure 5: The corrected ground-truth contract on the training-time development subset. A row is a positive only when `is_vessel` is true at HIGH or MEDIUM confidence; HIGH/MEDIUM non-vessels are background, and LOW-confidence rows become ignore regions. Every H100 cell trained and evaluated under this contract.

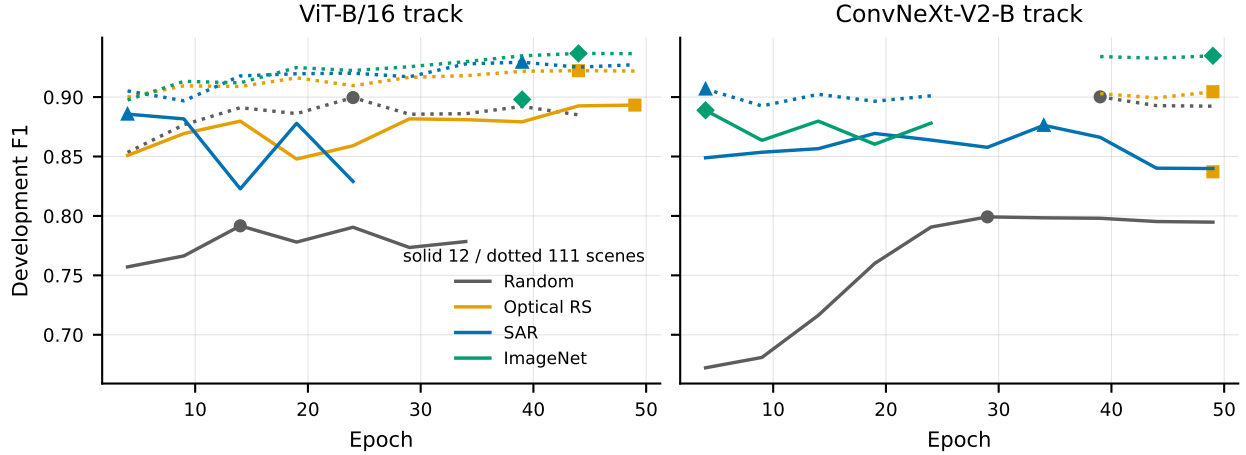


Figure 6: Development-F1 evaluation history for the smallest (12 scenes, solid) and largest (111 scenes, dotted) budgets. A marker identifies each selected checkpoint. The 12-scene ViT-ImageNet and CNN-optical curves were not saved, so only their markers appear. Curves that start late belong to resumed runs, which kept only post-resume evaluations. The SAR and ImageNet CNN arms separate from the random floor in the first evaluations, and checkpoint binding retains peaks that occur before the last evaluation.

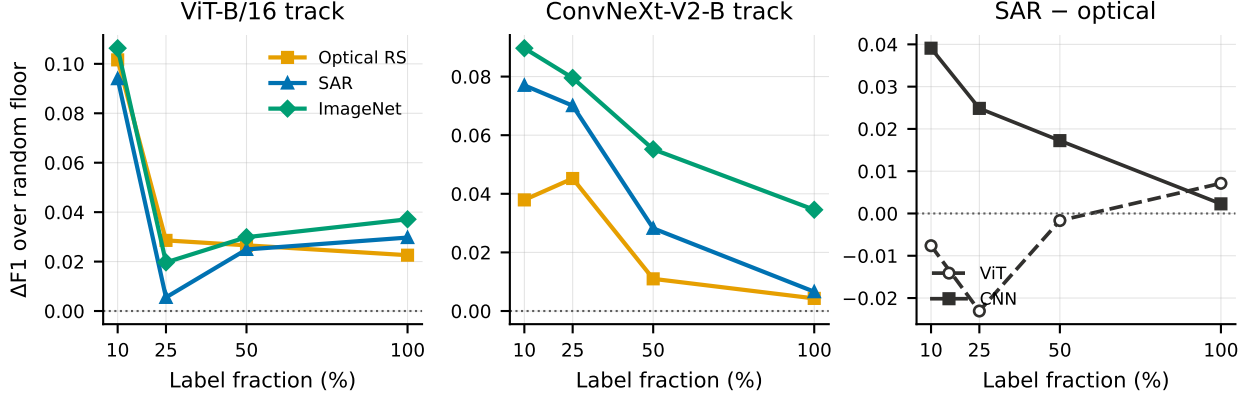


Figure 7: Left and center: transfer gain over the track-matched random floor (Eq. (2)) at each budget. Every gain is positive. Right: the SAR-optical contrast (Eq. (3)). It is positive for the CNN at every budget and shrinks toward zero at full data, but negative for the ViT below full data, so the modality effect depends on the architecture.

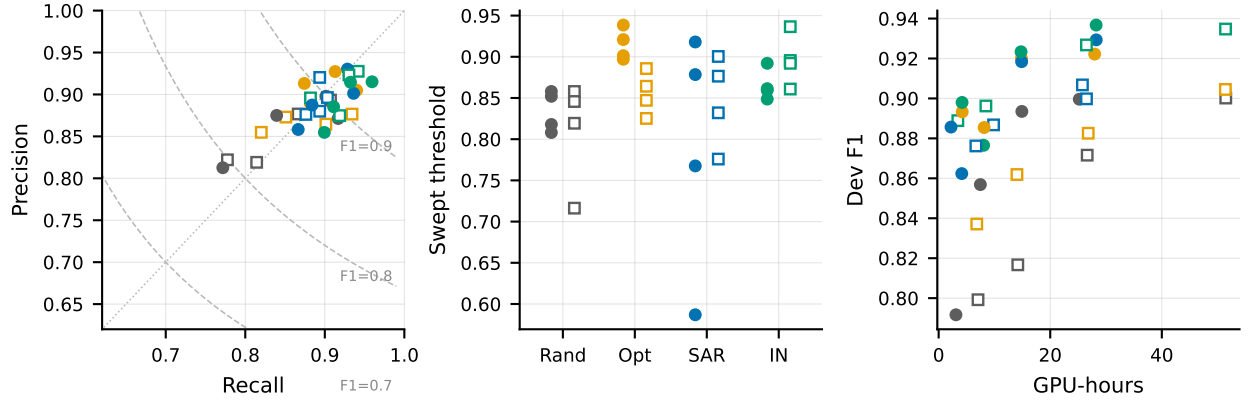


Figure 8: Operating points for all 32 cells. Filled circles show ViT, open squares show CNN, and colors identify roles as in Fig. 1. The left panel shows precision-recall positions with iso-F1 contours. The center panel shows the swept threshold for each role. The right panel shows development F1 against measured GPU-hours per cell.

C Extended Analysis

Operating thresholds and calibration. Each cell selects its operating point on the training-time development subset, and the cohort binds that point to the selected checkpoint. Figure 8 shows all 32 of them. The recorded thresholds vary widely (center panel): from 0.587 to 0.938 across the ViT cells and from 0.716 to 0.937 across the CNN cells. A single shared threshold would therefore change the operating point for most arms. At the bound operating point, precision exceeds recall in 14 of the 32 cells (left panel).

Training dynamics and cost. Figure 6 traces the development-F1 evaluation history for the smallest and largest budgets. The SAR and ImageNet CNN arms separate from the random floor in the first evaluations, and the ViT arms begin closer together. The 12-scene CNN-optical curve was not saved, but its selected checkpoint sits well above the random floor. Early stopping, with a patience of four development evaluations, ended 9 of the 32 cells before the 50-epoch cap. The median best-checkpoint epoch is 34 for the ViT track and 34 for the CNN track. Fine-tuning cost also differs between families. A ViT cell averaged 13.2 GPU-hours, and a CNN cell averaged 22.3 GPU-hours. The full matrix used 567.6 GPU-hours. Within each track, cost scales with the label fraction, so the scarce-label comparisons also require the least fine-tuning compute (Fig. 8, right).

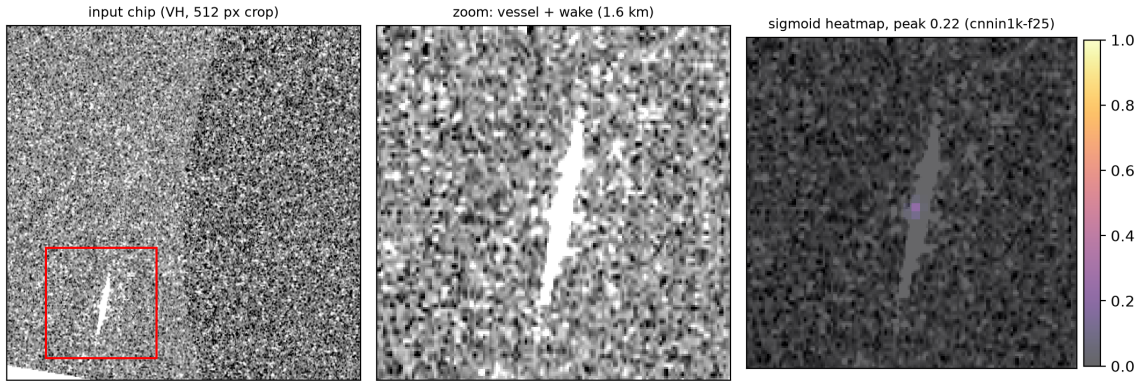


Figure 9: A vessel-and-wake target and the shared detector’s heatmap response from a development-phase probe on this study’s frozen splits (peak score 0.22). This sample is illustrative and not scored.

Qualitative example. Figure 9 shows a detector response on a development scene. The heatmap places a 0.22 peak on a clear vessel-and-wake target. This example illustrates why candidate generation uses a 0.05 floor before the per-cell threshold sweep. A fixed high candidate threshold would drop this detection. The panel is illustrative and contributes no scored evidence to the findings in Sect. 5.

D Reproducibility and Audit Record

The evidence tree records the frozen training cohort, each cell’s hash-bound completion marker, its full training curve, a sanitized runtime receipt (H100 hardware class, source SHA, strict-FP32 backend state), and the ground-truth audit receipt that binds every support count in Table 1 to the frozen split file. The generator validates marker-to-cohort hashes, metric consistency, and curve agreement before it writes a manuscript artifact. Two cells, 12-scene ViT-ImageNet and 12-scene CNN-optical, ran their last development evaluation in a resume that added no training step and did not save the curve. Their markers hash-bind the recovery record, and the generator accepts the missing curve only when that record selects exactly the marker’s checkpoint and result. Each human-verified result must hash to its entry in the final-evaluation completion record and bind the cohort, the cell’s test result, and its threshold; the generator admits all 32 or none. Checkpoint bytes stay outside the repository. We publish their SHA-256 bindings for archive verification. The submission excludes imagery, labels, weights, checkpoints, run trees, credentials, and private paths.

The detector uses 512-pixel crops, a stride-4 heatmap, Gaussian standard deviation two output pixels, a candidate floor of 0.05, and 120 m distance NMS. Training uses at most 50 epochs with development

evaluation every five epochs. The project fixes seed 0 and reports point estimates without uncertainty bands.